



Arrow Ballistic Simulator

User Manual

Version 9.0 — Adaptive RK4 — 2025/2026

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1. Introduction

The **Ballistic Simulator** is an advanced tool for analyzing arrow flights. It uses a **Runge-Kutta 4th-order (RK4) with adaptive steps** to calculate precise trajectories, considering aerodynamic drag, wind, atmospheric conditions, archer biometry, arrow characteristics, and sight system geometry.

In version 9 the simulator is fully internationalized (5 languages) and supports metric and imperial measurement systems.

2. Requirements and installation

The simulator is available as an open-source HTML files on GitHub: github.com/CP52/Arrow-ballistic-simulator-for-Archers.

How to use it:

🌐 **Online (recommended)**

Open directly in your browser: <https://arrow-sim.netlify.app>

Works on PC, Mac, iPhone, Android and tablet — no installation required.

3. User interface

3.1 Header and global controls

The program header contains four controls:

- 🌙 / ☀️ **Light/dark theme** (top-right corner): reduces eye strain in low-light conditions.
- **Language selection** (top-left corner): IT GB ES FR DE — all texts and labels update instantly.
- **m / yd** (top-right corner, next to the theme toggle): toggles between metric and imperial. All input fields and results convert automatically. Internal calculations are always in SI.

The main area is divided into two columns:

- **Left column:** expandable/collapsible parameter panels (arrow, bow, target, biometrics, environment, sight geometry).
- **Right column:** results area with metrics, interactive charts and sight table.

3.2 Arrow Panel

Parameter	Unit	Description
Mass	g / gr	Total weight of the assembled arrow (tip, shaft, nock, fletching)
Length	m / in	Total length from nock to tip
Diameter	mm / in	Shaft outer diameter
Spine	—	Shaft stiffness (e.g. 400, 500, 700). Affects the C_D factor.
Tip weight	g / gr	Weight of the insert/tip (affects FOC calculation)
Balance point	m / in	Distance from nock to balance point (for FOC calculation)
Tip type	selection	Standard / Broadhead / Blunt (affects aerodynamic drag)
Fletching	selection	Type and size of vanes (affects stability and C_D)

FOC (Front of Center): calculated automatically from the balance point. Optimal between 7–15%; values above 10% slightly increase drag but improve stability. Displayed in green in the info-box below the Target panel.

3.3 Bow Panel

Parameter	Unit	Description
Draw force	lb	Draw weight of the bow at its full draw length
Draw length	m / in	Draw distance from rest position
Brace height	m / in	String-to-grip distance at rest
Nocking offset	mm (fixed)	Vertical offset of the nocking point relative to the perpendicular to the string
Bow type	selection	Longbow / Recurve / Compound (automatically sets efficiency)
Efficiency	0–1	Energy Output (Auto: Longbow ~0.73, Recurve ~0.84, Compound ~0.89)

Nocking offset: the field is **always in mm** regardless of the selected measurement system (metric/imperial). The – and + increase/decrease by 1 mm. Typical values: 3–8 mm above the perpendicular. The simulator internally converts the value to angular deviation via arrow geometry (atan2) and subtracts it from the aiming angle.

3.4 Target Panel

Parameter	Unit	Description
Distance	m / yd	Horizontal distance to target center
Height	m / in	Height of target center relative to the archer's feet (negative = downhill shot)
Measured v_0	m/s o ft/s	If the checkbox is activated, the calculated speed will be overwritten with the one measured by the chronograph.

Info-box: below the v_0 checkbox the calculated speed and current FOC are displayed.

3.5 Archer Biometry Panel

Parameter	Unit	Description
Neutral launch height	m / in	Height of the arrow tip with horizontal posture (flat T)
Hip height	m / in	Rotation center of the torso during T-form rotation (approximately at the navel)
Shoulder-to-anchor	m / in	Effective T-arm: distance from shoulder to anchor point
Correzione posturale	flag	When active, iteratively recalculates the launch height as the shooting angle varies

Postural correction: when the archer rotates the T upward or downward, the arrow departs from a slightly different height than the neutral one. With this flag active the simulator iteratively converges (8 steps) to a consistent angle/height pair. The correction delta is shown in the results panel.

3.6 Environment Panel

Parameter	Unit	Description
Wind	m/s o ft/s	Positive = tailwind, negative = headwind
Temperature	°C or °F	Affects air density
Pressure	hPa o inHg	Local atmospheric pressure
Humidity	%	Relative humidity (CIPM-2007 formula)
Altitude	m o ft	Automatically corrects pressure and temperature via ISA model

3.7 Sight Geometry Panel

Parameter	Unit	Description
Eye–nock	m / in	Vertical distance between the eye and the nock when aiming at full draw. Enter in centimeters. Mediterranean grip: typically 10-15 cm. Stringwalking: typically 2-3 cm. Affects the projection in the Sight tab, not the calculations in the Stringwalking tab.
Nock–riser	m / in	Horizontal distance from nock to riser/sight plane
Min. dist.	m / yd	Minimum distance in the sight table
Max. dist.	m / yd	Maximum distance in the sight table
Step	m / yd	Interval between distances in the table

4. Running the simulation

Press the  SIMULATE button (or equivalent in the selected language) to start the calculation. The simulator:

- searches for the optimal angle with a whole-degree search, then refines by bisection to $\pm 0.01^\circ$;
- integrates the trajectory with adaptive RK4 (tolerance 10^{-6} m);
- calculates the maximum range and corresponding optimal angle;
- generates the sight table for all requested distances.

5. Interpreting the results

5.1 Main metrics

Six cards in the upper part of the results area:

Metric	Meaning
Aiming angle	Optimal launch angle (already corrected for nocking offset and posture)
v_0	Initial arrow speed (calculated or measured by chronograph)
Tempo	Flight time to target
H max	Maximum height reached by the trajectory
Drop	Arrow drop relative to the line of sight at target distance
Gittata max	Distanza massima raggiungibile con relativo angolo ottimale (mostrato in gradi)

Corrections panel: below the cards, when active, the nocking offset delta (effective angle vs. aiming angle) and the postural correction in centimeters of height are displayed.

5.2 Interactive charts

Four tabs, all with interactive tooltip on mouse-over:

-  **Trajectory:** real curve (with drag), ideal (without drag) and line of sight. Orange circle (target) and green (impact point).
-  **Speed:** total speed, horizontal component (Vx) and vertical component (Vy) along the path.
-  **Energy:** percentage of residual kinetic energy relative to launch.
-  **Drop:** drop in cm (or in) for each distance relative to the line of sight.

5.3 Sight Tab

Sight table

For each configured distance, the table shows:

- **Drop (cm / in):** arrow drop relative to the line of sight.
- **Riser (cm / in):** geometric projection onto the riser plane (position of the sight mark).
- **E (J or ft·lb):** residual kinetic energy at impact.
- **V (m/s or ft/s):** residual speed at impact.

The row corresponding to the target distance is highlighted in yellow. The visual scale on the right of the panel graphically shows the mark positions. The green dot indicates the 30 m projection of the ideal laser beam, which practically corresponds to the eye–nock distance for distances > 20 m.

5.4 Stringwalking Tab

The row corresponding to the target distance is highlighted in yellow. The visual scale on the right of the panel graphically shows the mark positions. The green dot indicates the 30 m projection of the ideal laser beam, which practically corresponds to the eye–nock distance for distances > 20 m.

The Stringwalking tab is for archers who practice stringwalking, a technique in which the finger position on the string varies depending on the distance to the target.

The table shows, for each distance configured in the Sight Geometry panel, how many millimeters must be lowered on the string from the nock to hit the target at the height indicated in the Target panel.

Table columns:

- **Dist** (m/yd) Distance to the target
- **Hook** (mm) Millimeters below the nock. Positive value = lowered on the string. Negative Value = correct aiming (rare, only at very close ranges).
- **Sight Angle** (°) Launch angle required to hit the target.
- **E** (J/ft lb) Residual kinetic energy at impact.
- **V** (m/s/ft/s) Residual velocity at impact.

Physics of calculation: Stringwalking is equivalent to dynamically varying the nocking offset: lowering the grip by X mm/cm on the string is equivalent to giving the same "lift" to the arrow as would be obtained by lowering the nocking point by the same amount.

The simulator exploits this equivalence: for each distance, it finds the optimal launch angle using RK4 integration, then calculates the grip point with the formula:

$\text{gripmm} = \tan(\text{launch_angle}) \times (\text{arrow_length} \times 0.9) \times 1000$ where $\text{arrow_length} \times 0.9$ is the approximate distance between the rest and the nock (restToCock).

In other words: the archer physically tilts the entire system (bow + arrow) pointing the arrowhead (aligned with the eye) on the target, and the grip point is the position on the string that will result in the correct lift angle.

Practical notes: The 0 mm reference corresponds to the grip point immediately below the nock. Each archer can adapt the values to their personal reference by subtracting the offset of their usual grip. — Recalculate after each change to the arrow or bow parameters. — The row corresponding to the target distance is highlighted in yellow.

6. Data export

6.1 Export CSV

The  Export CSV button generates a text file with the complete sight table, compatible with Excel, LibreOffice Calc and any spreadsheet application. It can be copied to the clipboard or downloaded as a file.

6.2 Export PDF (Sight Scale)

The  Export PDF button generates an A4 page containing:

- **Header:** title in pure ASCII, v_0 , aiming angle, target distance, date.
- **Complete sight table:** with target distance highlighting and alternating rows.
- **1:1 visual scale:** the position of the marks on the riser at actual size.
- **Two 5 cm calibration rulers: one vertical (to the left of the scale zero) and one horizontal (to the right). Overlay a physical ruler to verify that the print was executed at 100% without automatic scaling — a setting to verify in the printer driver.**

 **Be careful when printing scale: many printers automatically scale the document to 95-98%. Always check the calibration rulers before transferring the markings to the media.**

7. Practical tips

- **Chronograph:** enter the measured speed to maximize simulation accuracy.
- **FOC:** keep between 10–15% for the best trade-off. Calculated automatically from the balance point.
- **Downhill shots:** set negative heights to simulate shots from a hide or elevated position.
- **Kneeling shots:** adjust Neutral launch height and Hip height.
- **Environmental conditions:** update temperature, pressure and wind for simulations in real competition conditions.
- **PDF:** always check the calibration rulers before using the scale on the riser.
- **Offline:** after the first load the simulator works without a connection. Ideal in the field.
- **Stringwalking:** set the Eye-to-nock field to 2-3 cm before using the Stringwalking tab. The Grip point (mm) column gives the string position for each distance.

8. Technical notes

The simulator implements the following physical models:

- **Adaptive RK4:** step control via Richardson extrapolation, tolerance 10^{-6} m.
- **Drag model:** Reynolds-number-dependent C_D with 7 transition zones; corrected for FOC and fletching type. Data from JAXA/MSBS (Ortiz Enriquez, UEC Tokyo 2021).
- **Atmosphere:** density with CIPM-2007 formula; altitude correction with ISA thermal gradient.
- **Sight projection:** full parallax geometry, not the eye–nock $\approx$ constant approximation.
- **Nocking offset:** mm $\rightarrow$ angular deviation conversion via $\arctan$ on arrow geometry.
- **Stringwalking:** grip point calculated by equivalence with the nocking point offset: $\text{grip_mm} = \tan(\alpha) \times \text{restToCock} \times 1000$, where α is the RK4 optimal angle for the given distance.
- **No build step: Pure HTML + React (CDN) + jsPDF (CDN).** ~1,800 lines, single self-contained file.

9. Limitations

- **2D only:** no lateral drift from crosswind or spine effects.
- **Rigid arrow:** does not model shaft flexion or the archer's paradox.
- **Constant efficiency:** does not use the real bow load curve.
- **No Magnus effect:** negligible below ~1,500 RPM (field and 3D).
- **Simplified stringwalking:** the model does not account for spine effects or the residual vertical component of the string force resultant. For the angles typical of field shooting the approximation is negligible.

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